

Shabad Vaswani

shabad.vaswani@nyu.edu | (347)-651-6352 | linkedin.com/in/shabadvaswani | github.com/shabadvaswani | New York, NY | Portfolio

SUMMARY

AI/full-stack software engineer shipping production LLM agents, MCP tools, RAG, Python APIs, React/TypeScript products, and streaming ETL. Strong in AWS/GCP, PostgreSQL, MongoDB, LangChain, observability, and end-to-end delivery from design through deployment.

EDUCATION

New York University

May 2026

Master of Science, Computer Engineering (Merit Scholarship)

GPA: 3.7/4.0

Relevant Coursework: MLOps, Big Data Management, Machine Learning, Deep Learning, Distributed Systems, Network Security.

Ganpat University

May 2024

Bachelor of Technology, Computer Engineering

India

Relevant Coursework: Database Systems, Computer Networks, Operating Systems, Data Structures and Algorithms, Artificial Intelligence.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Go, Java, C++

AI Agents & Automation: MCP, LangChain, multi-agent orchestration, agentic AI, tool use, RAG, GraphRAG, structured outputs

Full Stack & Data: React, Next.js, FastAPI, Django, REST APIs, PostgreSQL, MongoDB, Redis, Kafka, Spark, ETL

Cloud & Reliability: AWS, GCP, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Backend Developer, IAppSolutions

January 2026 – April 2026

Remote

- Built a low-latency Python tool layer for two production voice agents, exposing six Google Calendar operations for 30–60 daily bookings.
- Designed multi-tenant PostgreSQL routing for OAuth credentials and Calendar IDs, maintaining sub-350 ms latency.

Backend Developer, OddlyOptic

September 2025 – December 2025

New York, NY (Remote)

- Built production multi-agent services with Python/FastAPI and Docker across AWS and GCP, orchestrating OpenAI, Claude, and provider-specific models behind unified APIs.
- Developed MCP servers and structured tool interfaces for provider abstraction, delegated workflows, and reusable agent integrations.
- Built RAG pipelines over Twitter/X, Reddit, and web sources; added Prometheus observability and staged A/B rollouts for safer releases.
- Implemented JWT authentication and automated credential handling through AWS and GCP secret-management systems.

Python Backend Developer, Tiny Archives

June 2025 – August 2025

Oxnard, CA (Remote)

- Built Django, React, and PostgreSQL features for metadata-rich workflows and database-backed PDF/HTML exports through REST APIs.
- Optimized PostgreSQL queries and reporting pipelines; improved reliability through testing, debugging, pull requests, and code reviews.

Project Trainee, Arrow Company

January 2024 – May 2024

Ahmedabad, India

- Automated Python financial data pipelines and scheduled jobs, cutting manual tracking effort by 80% and improving reliability.

SELECTED PROJECTS

STEWARD – Agentic Portfolio Manager: *Go | GraphRAG | Neo4j | Vector Retrieval | Docker*

- Engineered a four-stage autonomous portfolio workflow using LLM agents, GraphRAG, a knowledge graph, and vector retrieval.
- Built a concurrent Go backend with strict JSON validation to prevent malformed model outputs from propagating through agent workflows.

AEON – Streaming ETL and Analytics Platform: *Python | Kafka | Spark | Cassandra | FastAPI*

- Built Kafka/Spark ETL for 40,971 minute-level network, biometric, and environmental records across 30 features.
- Designed Cassandra time-series storage and FastAPI services for scalable ingestion, historical retrieval, and operational analytics.

Egocentric World Memory: *Python | OpenCLIP | NeRF | Semantic Retrieval | Spatial Evaluation*

- Built pose-aligned semantic memory from real and NeRF observations, linking language-relevant views to 3D camera regions.
- Evaluated spatial retrieval at K=3, 5, and 10 against random baselines; identified NeRF smoothing as a localization failure.

MedAI – ML Training and Serving Platform: *Python | FastAPI | MLflow | Kubernetes | GitHub Actions*

- Deployed Dockerized FastAPI inference services on Kubernetes and tracked experiments, models, and evaluation metrics with MLflow.
- Built CI/CD automation for tests, container builds, deployment validation, and reproducible model-service releases.

CLOUD CREDENTIALS

Google Cloud Skills Boost: Secure Cloud Network; App Development Environment; Compute Engine Load Balancing; ACE Exam Prep Study Guide.